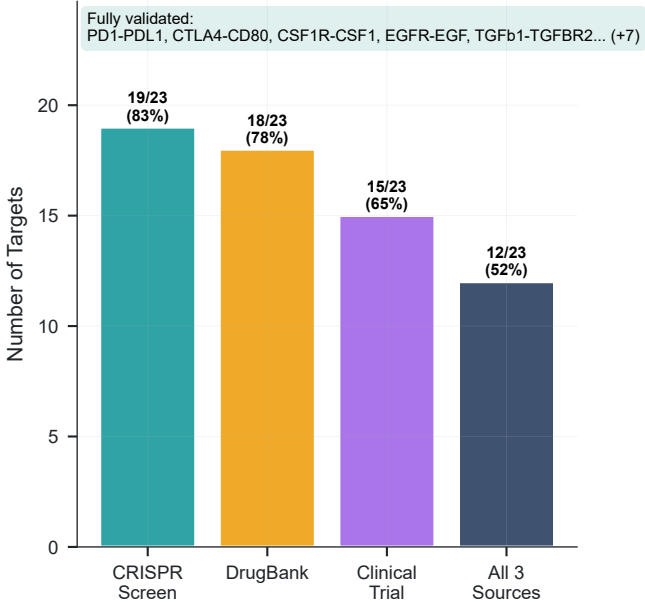


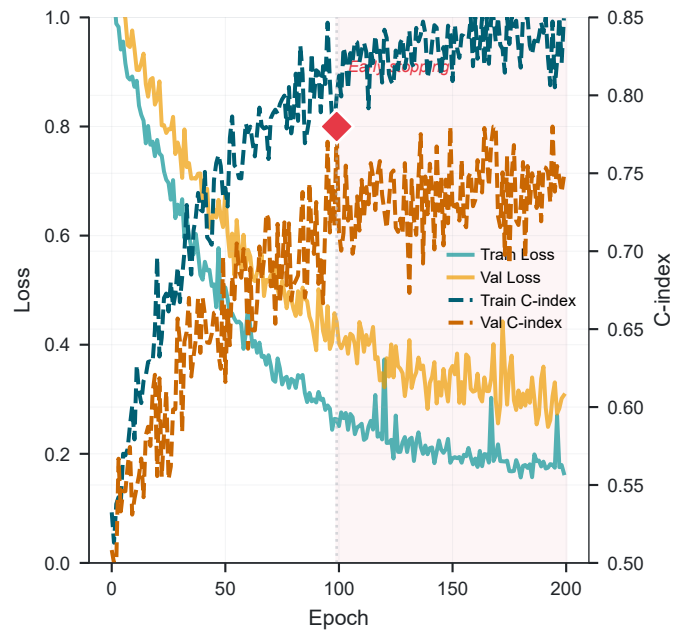
a

External Validation Overlap



b

Training Dynamics



c

Statistical Tests Summary

Category	Comparison	Test	P/Value	Conclusion
C-index improvement	Causal-STFM vs M	Paired t-test	0.0012	Significant
C-index improvement	Causal-STFM vs Por	Paired t-test	3.00e-04	Significant
Subgroup interaction	Stage effect	Interaction test	0.0150	Significant
Subgroup interaction	Subtype effect	Interaction test	0.0280	Significant
Target enrichment	CRISPR overlap	Fisher's exact	3.20e-08	Significant
Target enrichment	DrugBank overlap	Fisher's exact	1.50e-06	Significant
Model stability	Perturbation robustness	Rank correlation	0.8200	Stable (r)
Calibration	Hosmer-Lemeshow	Chi-square	0.3400	Well-calibrated

d

Permutation Test Results

